SME Technology Development (R&D) Support Project Research and Development Plan

Project Overview

Project Title	Development of an RAG-based Business-Specific Real-Time Multilingual Translation AI Solution Aimed at Revolutionizing SME Export Sales through a Global Sales, Consultation, Interpretation, and Document Automation Integrated Platform
Development Goals	Implement a business-specific AI model. Develop an RAG-based multilingual translation model and a real-time streaming speech interpretation model (within seconds delay, supporting more than 8 languages) that learns company-specific knowledge such as product names, technical specifications, and prices. Develop agents for export process automation throughout the entire lifecycle, including automatically generating export documents like quotations and product pages. Establish a platform integrating unique regulatory certification formatting (eCTD), KC, CE, FCC, etc. to eliminate global entry barriers for SMEs and achieve sales innovation.
Development Method	Establish a critical differentiation by utilizing a proprietary corpus of 1.1 million sentence pairs for business, 600,000 for regulatory certification, and 500,000 exclusive corpora, which cannot be easily replicated in a short term, for domain-specific fine-tuning, thereby creating an overwhelming market entry barrier. [For AI model enhancement] Combine the RAG technology, which injects company-specific knowledge into a hybrid language model based on NLLB/EXAONE, to reduce hallucination phenomena by 70%. Secure simultaneous interpretation performance within seconds delay utilizing InfiniSST and Whisper-based technology. [Commercialization of SaaS Platform] Develop a cloud SaaS web, mobile, API integrated environment that allows end-users to integrate and use their data within 3 days without any IT workforce by leveraging the verified AI platform (Oz-i) infrastructure of the lead organization.

Research Team Composition	[Lead] JPCN Co., Ltd. (Platform/Commercialization oversight): An AI company with 25 years of operation (revenue of 27 billion KRW) possessing the Oz-i AI platform and diverse know-how in R&D projects, ensuring the construction of SaaS platforms, API integration, and the commercialization of cloud infrastructure. [Cooperation①] StunningBox Co., Ltd. (AI Model/Data): An AI translation data company with 15 years of expertise accumulating specialized datasets, responsible for AI translation fine-tuning, provision of the core asset exclusive corpora, and multi-agent regulatory certification formatting technologies. [Cooperation②] 3 Demand Companies (Field Proof Validation): Sunlight Illucom Co., Ltd. (auto parts), GenuineGrip Co., Ltd. (specialized footwear), and Gray Whale (K-Beauty) that are award-winning companies, will provide exemplified data by category and quantitatively verify effects through A/B testing. .
Expected Effects	[Economic/Industrial Impact] Enhance export order rates of demand companies by more than 20% and significantly reduce the processing period for regulatory certification documents from the current 4-8 weeks to 1-2 weeks. By the 5th year of commercialization, create revenue of 22 billion KRW (cumulative 68 billion KRW over 5 years), and generate 51 new jobs. Distribute to over 500 SMEs via AI vouchers and KOTRA linkages. Achieve a cumulative export substitution effect of 44 billion KRW over 5 years. [Securing World-Class Technology] Achieve translation accuracy of BLEU 65+, speech interpretation delay within 2 seconds, and document structure retention rate (TEDS) over 70%, capturing the B2B domain-specific translation market that global big-tech (Google·DeepL) could not solve.

Self-Diagnosis

Project Title	Global Sales, Consultation, Interpretation, and Document Automation Integrated Platform for SME Export Sales Innovation	
Checklist	Applicable	Not Applicable
1. Does the current applicant (research and development organization, etc.), hereinafter referred to as 'the applicant', not meet the eligibility criteria set forth in the project announcement, etc.?		✓
2. Are there any related documents, set out in the project announcement, etc., that the current applicant has not submitted?		✓
3. Is the current applicant subject to participation restrictions in national R&D projects?		✓
4. Has the applicant ever been sanctioned for misuse of labor costs or false transactions, such as being restricted from participating in government R&D?		✓
5. Is the applicant currently bankrupt or a business that is closed or suspended?		✓
6. Is the applicant currently subject to any tax delinquencies such as national taxes, local taxes, or customs duties by the tax authorities, or has information such as fines in arrears, industrial accident or employment insurance arrears been registered with the Comprehensive Credit Information Center?		✓
7. Is the applicant currently listed in the registry for debt non-compliance under the Civil Execution Act or registered as a defaulter with the Comprehensive Credit Information Center?		✓
8. Has the applicant had a bankruptcy, rehabilitation process, or personal rehabilitation process filed with the court?		✓
9. Based on the recent fiscal year's closing, does the applicant have a debt ratio of 1,000% or more or is in a state of total capital erosion?		✓
As a result of the self-assessment, does the proposed project fall under the security project classification criteria outlined in Article 45, Paragraph 1 of the Enforcement Decree of the Innovation Act?		✓

※ In addition to the matters specified in the above self-diagnosis table, applicant eligibility requirements as stipulated in the project announcement and related regulations

The application may be excluded from support or the selection canceled based on the review results.

※ The criteria for 'currently' in the above confirmation matters are based on the deadline and other requirements specified in the project announcement and related regulations.

This is the specified point for confirmation.

※ Prepared by each host and joint institution

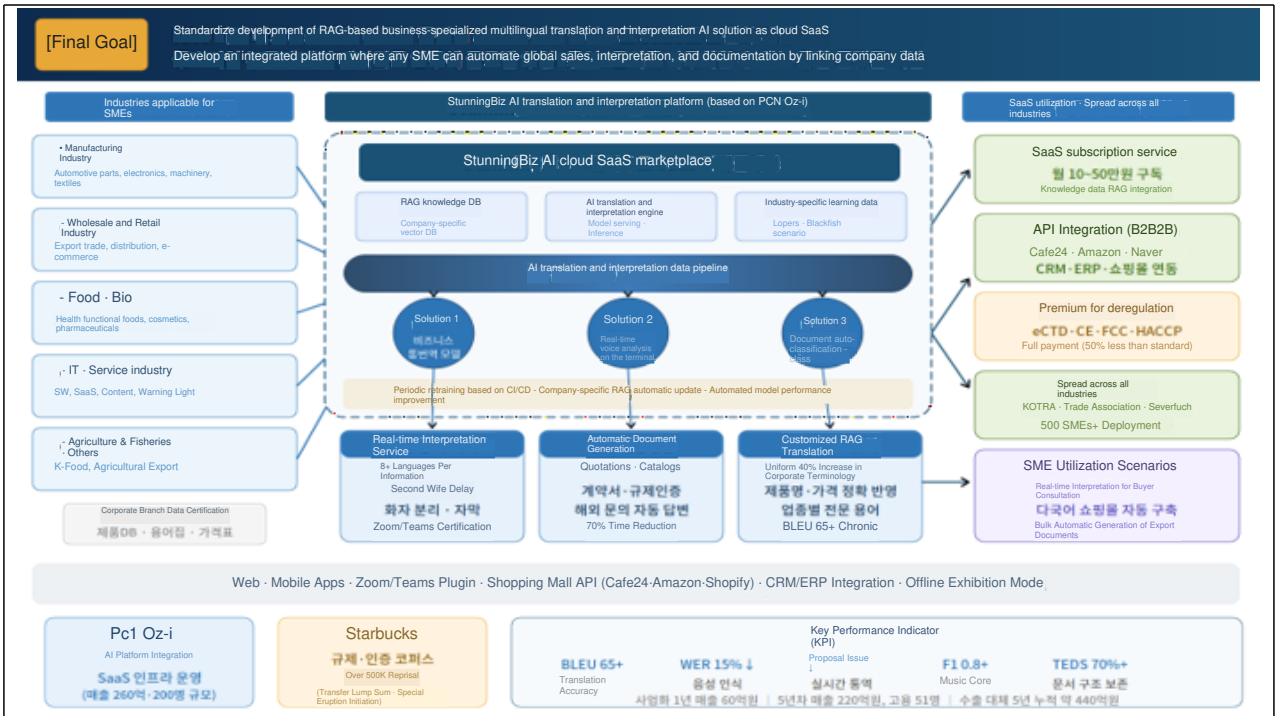
The applicant has verified the above matters, and if there is any discrepancy with the facts there is no objection to potential exclusion from support or cancellation of selection.

Year 2026 July 01 Day

Applicant for the host organization: Eom Sang-hyun Seo Yang-mun

Name of the host organization corporation: (Inc.) PCN

1. Technology Development Goal



1) Business Promotion Background and Necessity

(1) Environmental Analysis

Export SMEs face complex barriers during overseas business activities, like translation not reflecting company-specific contexts, inappropriate interpretation structure for real-time consultations, and high-cost, lengthy regulatory certification document responses. Hence, a domain-specialized AI platform connecting the entire export business cycle based on corporate data and professional terminology beyond simple translation functions is needed.

Core Barriers	Field Issues	Need for Technology Development
① Business Context · Professional Terminology	General-purpose AI fails to sufficiently reflect product names, technical specifications, prices, and contract terms unique to each company, risking misinterpretation of core information during negotiations and contracts	Connect company-specific product DB, price list, and glossary with RAG for real-time incorporation into translation, interpretation, and document generation
② Real-time Interpretation Performance	The existing 3-step structure of STT→MT→TTS incurs a cumulative delay of 2-5 seconds or more, and has limitations in background noise, speaker separation, and professional terminology handling	Achieve interpretation within 2 seconds through E2E streaming architecture and speaker separation, professional terminology injection
③ Regulatory Certification Documents	CE, eCTD, FCC, and HACCP have intricate formats and structures per country, posing a significant burden of preparation time and outsourcing costs for SMEs	Automate translation, structuring, and formatting of regulatory documents and support submission document creation based on expert review (HITL)

(2) Market Growth and Urgency of Development

◦ The global business AI translation market is expected to grow from about 1.8 billion USD in 2025 to over 7 billion USD by 2030, with rapidly expanding demand for business domain specialization.